

ME 515: Carbon Materials and Technology

Course Code: ME 515

Course Name: Carbon Materials and Technology

L-T-P-C: 3-0-0-3

Prerequisite: None (basic knowledge of material science expected)

Students intended for: M.Tech. / M.S./ Ph.D./ B.Tech. (4th year)/ M.Sc.

Elective or Core: Elective

Approval: 28th Senate

Course contents

- **Introduction to carbon**

- Why should one study carbon materials and manufacturing?
- Carbon economy
- Atomic structure and hybridization
- Carbon allotropes
- Nomenclature and terminology [6 Lectures]

- **Microstructure of carbon materials**

- a Bulk industrial carbon

- Graphite: natural and pyrolytic
 - Activated carbon
 - Glass-like carbon
 - Granular amorphous carbon

- b Carbon fibers and composites

- Activated carbon fiber
 - Carbon fiber reinforced plastics

- c Carbon nanomaterials

- Graphene
 - Carbon Nanotube
 - Fullerene
 - Graphite whiskers
 - Diamond-like carbon

- d Characterization of carbon materials (discussed in a-c) by X-Ray Diffraction, Raman spectroscopy and electron microscopy[15 Lectures]

- **Raw materials**

- Polymer precursors (Polyacrylonitrile, cellulose, resins, PVC etc.)
 - Needle coke (petroleum and pitch based)
 - Coal and its distillation
 - Gaseous hydrocarbons (for CVD)

- Carbon yield and mechanism of carbonization (thermodynamic and kinetic aspects) [4 Lectures]

- **Properties of carbon materials**

- Crystallinity and electrical conductivity
- Electrochemistry and surface chemistry
- Mechanical and thermal properties [4 Lectures]

- **Manufacturing techniques for carbon materials**

- Pyrolysis (activated carbon, glassy carbon, pyrolytic graphite)
- Electrospinning (carbon fibers)
- Chemical vapor deposition (graphene and CNT)
- Composite preparation (material/ binder interface)
- Mechanical property testing methods [4 Lectures]

- **Applications**

- Graphite electrodes
- Carbon-fiber composites in automobile industry
- Carbon-based micro and nano devices (sensors, microelectrodes etc.)
- Filters and adsorbers [6 Lectures]

- **Special topics**

- Health and environmental safety of carbon nanomaterials
- Carbon-based flexible electronics
- Future of carbon technology [3 Lectures]

Text Books:

1. Timothy D. Burchell, **Carbon Materials for Advanced Technologies**, Elsevier, 1999.

References:

1. Jenkins, G. M. & Kawamura, K., **Polymeric carbons—carbon fibre, glass and char**, Cambridge University Press, 1976).
2. Marsh, H. & Rodríguez-Reinoso, F., **Activated carbon**, Elsevier, 2006.
3. Kong, L. B., **Carbon nanomaterials based on graphene nanosheets**, CRC Press, 2017.
4. Chung, D.D.L. **Carbon fiber composites**, Butterworth-Heinemann, 1994